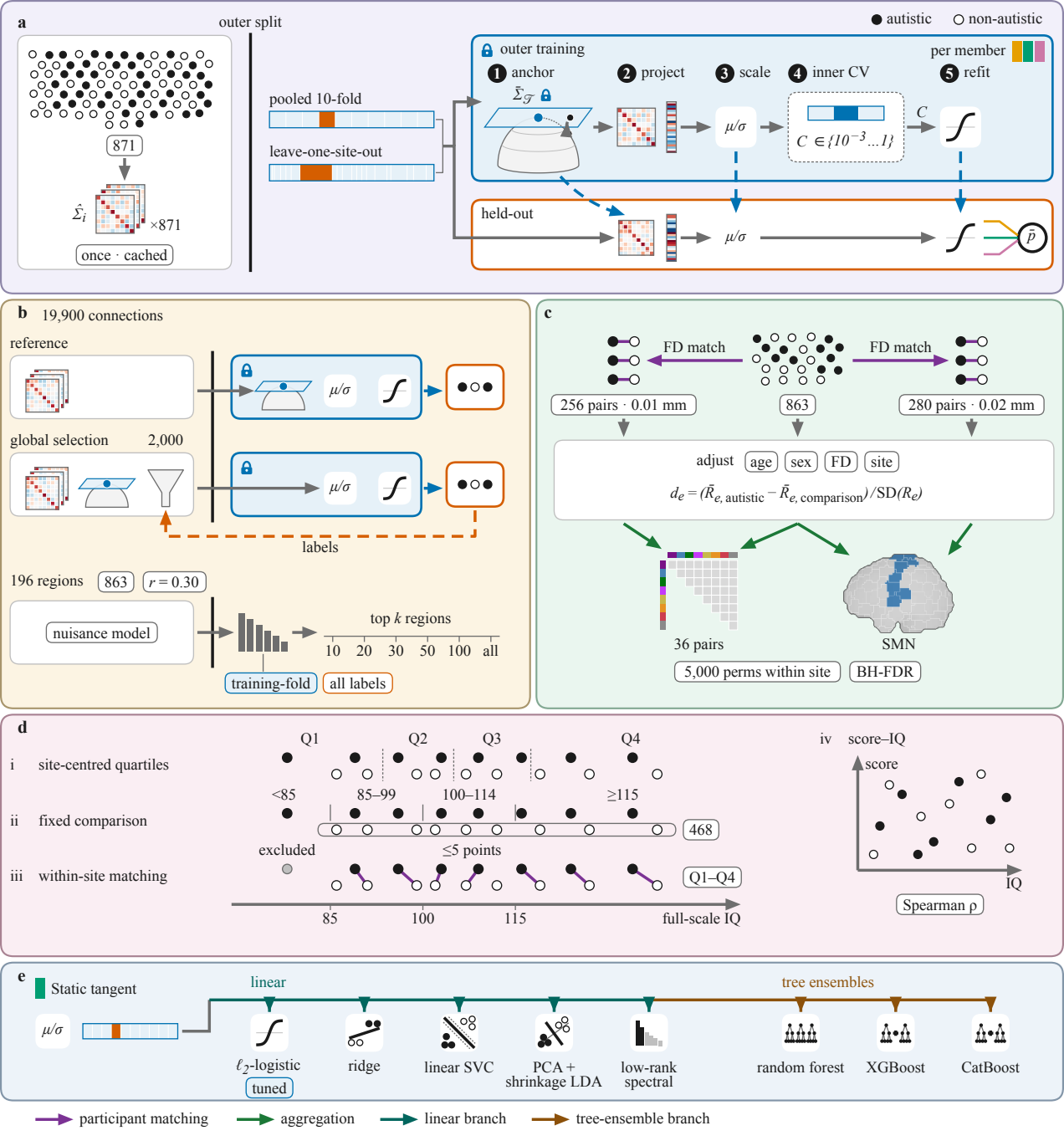

a
outer split
autistic
non-autistic
871
$\hat{\Sigma}_i$
×871
once · cached
pooled 10-fold
leave-one-site-out
outer training
per member
1 anchor
2 project
3 scale
4 inner CV
5 refit
$\bar{\Sigma}_{\mathcal{T}}$
$\mu/\sigma$
$C \in \{10^{-3} \ldots 1\}$
$C$
held-out
$\bar{p}$
b
19,900 connections
reference
global selection
2,000
labels
196 regions
863
$r = 0.30$
nuisance model
top $k$ regions
10 20 30 50 100 all
training-fold
all labels
c
FD match
FD match
256 pairs · 0.01 mm
863
280 pairs · 0.02 mm
adjust
age
sex
FD
site
$d_e = (\bar{R}_{e,\,\text{autistic}} - \bar{R}_{e,\,\text{comparison}})/\mathrm{SD}(R_e)$
36 pairs
SMN
5,000 perms within site
BH-FDR
d
i site-centred quartiles
ii fixed comparison
iii within-site matching
Q1
Q2
Q3
Q4
<85
85–99
100–114
≥115
468
excluded
≤5 points
Q1–Q4
85
100
115
full-scale IQ
iv score–IQ
score
IQ
Spearman ρ
e
Static tangent
$\mu/\sigma$
linear
tree ensembles
$\ell_2$-logistic
tuned
ridge
linear SVC
PCA + shrinkage LDA
low-rank spectral
random forest
XGBoost
CatBoost
participant matching
aggregation
linear branch
tree-ensemble branch